MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER Value Addition Course (VAC) - MG3VACPES200 - SAFE TRAINING IN SPORTS (2024 ADMISSION ONWARDS)

QP Code -25900552

PART A

1. a) ACSM (American College of Sports Medicine)
2. b) Providing equal opportunity, privacy, and respect to all genders
3. b) Undiagnosed heart abnormalities
4. b) Medical screening and emergency preparedness
5. b) Clearly display rules, hazards, and emergency procedures

PART B

6. Physical Activity Readiness Questionnaire. It is a standard pre-exercise screening tool used to identify individuals who need medical clearance before starting an exercise program.
7. To ensure understanding, provide a permanent record, and reduce legal liability. Written communication removes doubt and demonstrates that the facility has made a reasonable effort to inform participants of risks and rules, which is a key part of risk management.

8. To provide immediate care in the event of an injury or medical emergency, which can significantly impact the outcome. Quick access to communication (like a phone) is crucial for activating Emergency Medical Services (EMS) as part of the Emergency Action Plan (EAP).

9. Proper flooring is critical for both safety and performance; non-slip surface to prevent injury and shock-absorbing rubber protects against falls and dropped weights, while a stable, firm surface like wood provides a solid base for efficient force transfer during lifts.

10. Hyperthermia is a condition of abnormally high body temperature caused by the body's inability to regulate heat. High environmental temperature and humidity, inadequate hydration, excessive clothing, lack of acclimatization, and intense or prolonged exercise are major conditions that can lead to heat-related illness during sports activities.

11. Voluntariness ensures that the participant agrees to take part in the activity without any coercion, pressure, or undue influence. It is a fundamental ethical principle that

respects the individual's independence and their right to make decisions about their own body and participation.

12. Key responsibilities include:

    - Providing direct and constant supervision.
    - Instructing and correcting exercise technique.
    - Ensuring safety (e.g., spotting, enforcing rules).
    - Monitoring the environment and equipment.
    - Being prepared to implement the Emergency Action Plan.

PART C

13. **Drinking Water:** Essential for maintaining hydration, regulating body temperature, and preventing heat-related illnesses, thus supporting both safety and performance.

    Restrooms & Showers: Promote hygiene, prevent the spread of infection, and provide convenience, which boosts athlete morale.

    Telephones: A critical component of the emergency communication plan for contacting EMS. The text specifies a phone should be accessible, including for persons in wheelchairs.

    First Aid Facilities: Enable the immediate and effective management of injuries, which is a fundamental standard of care and a core part of any Emergency Action Plan.

14. Preventive measures for Exertional Rhabdomyolysis include:

    - Gradual Progression: Slowly increasing the volume, intensity, and novelty of exercise, especially after a period of inactivity.
    - Adequate Hydration: Ensuring athletes are well-hydrated before, during, and after training.
    - Proper Conditioning: Building a solid fitness base before introducing highly intense workouts.
    - Education: Teaching athletes and coaches to recognize the early signs and symptoms (e.g., severe muscle pain, dark urine).

- Appropriate Workout Design: Avoiding extremely high-volume, repetitive exercises to failure, especially in unconditioned individuals.
- Acclimatization: Allowing time for athletes to adapt to environmental conditions like heat and humidity.

15. Condition: A serious medical condition characterized by the rapid breakdown of skeletal muscle tissue due to overexertion, releasing harmful contents like myoglobin into the bloodstream, which can lead to kidney damage.

    Causes: Intense, prolonged, or unfamiliar exercise; especially high-volume eccentric exercise; often combined with dehydration and heat stress.

    Warning Signs**:** Severe muscle pain, weakness, and swelling; dark brown or tea-coloured urine; decreased urine output.

    Preventive Measures: Gradual progression of training, ensuring proper hydration, adequate recovery, athlete and coach education, and appropriate workout design that considers environmental conditions.

16. **Location:**

    Floor Level: Ideally on the ground floor to support heavy loads and minimize noise disruption.

    Load-bearing Capacity: The floor must be stable enough to support heavy equipment (min. 100 lbs/sq ft).

    Proximity: Should be easily accessible to athletes but located away from offices/classrooms to avoid noise issues.

    Access Routes:

    Doorways: Double doors are necessary for moving large equipment in and out.

    Accessibility: Must be accessible for people with disabilities, requiring ramps or lifts for any change in height.

    Traffic Flow: Design should allow for clear, unobstructed walkways (min. 36 inches wide) for safe movement and emergency way out.

17. Supporting facilities are vital for a holistic training environment.

Health: Locker rooms and showers allow athletes to clean up, reducing the risk of skin infections and promoting overall well-being.

Hygiene: Adequate cleaning schedules and supplies (germicidal cleaners, mops, vacuums) prevent the growth and spread of bacteria, mold, and fungi on surfaces and equipment.

Safety: Proper storage rooms for equipment and cleaning supplies reduce clutter and tripping hazards. Access to phones and first aid kits is directly related to emergency response safety. Good lighting and ventilation create a safe and comfortable training atmosphere.

18. **Signs & Symptoms:**

    a. **Severe Muscle Pain, Weakness, and Swelling:** Caused by the inflammation and breakdown of muscle cells.

    b. **Dark Brown or Tea-Colored Urine:** Caused by the presence of myoglobin, a muscle protein, being released into the bloodstream and filtered by the kidneys.

    c. **Decreased Urine Output:** Occurs if myoglobin causes damage and blockage in the kidneys, potentially leading to acute kidney failure.

Physiological Basis: Extreme exercise can cause tiny tears in muscle fibers. When the exercise is too intense or unfamiliar, these tears become severe, causing the muscle cells to rupture and die. When the muscle cells break down, they release their contents—including a protein called myoglobin—into the bloodstream. The kidneys then have to filter this debris, and the high levels of myoglobin can overwhelm and damage them, leading to serious complications.

PART D

19. The PAR-Q (Physical Activity Readiness Questionnaire) is a simple, cost-effective, and widely used first-step screening tool.

1. Purpose and Function: It is designed to identify individuals who have signs, symptoms, or risk factors for various cardiovascular, metabolic, or musculoskeletal diseases. It consists of a short list of "Yes/No" questions about an individual's health history and current symptoms.

2. Risk Stratification:

d. If the participant answers "No" to all questions, they are typically cleared to begin a moderate-intensity exercise program without needing to see a physician.

e. If they answer "Yes" to one or more questions, it signals a need for further medical evaluation and clearance from a healthcare provider before starting or intensifying an exercise program. This process effectively stratifies individuals into "low-risk" and "needs further screening" categories.

3. Legal and Ethical Benefits: Using the PAR-Q helps strength and conditioning professionals fulfill their "duty of care." It documents that a pre-participation screening was conducted, which is a key step in risk management and obtaining "informed consent." It protects both the athlete (by identifying potential health risks) and the professional (by demonstrating a standard of care was followed).

4. Limitation and Role: The PAR-Q is a screening tool, not a diagnostic tool. It does not replace a comprehensive medical exam but serves as an essential gatekeeper to ensure that individuals at potential risk are directed to the appropriate medical professionals.

20. The role of a PE teacher is to act as a first responder, ensuring the immediate safety and care of the injured student while managing the overall situation.

- 1. Primary Responsibilities:
    - Ensure Scene Safety: Quickly assess the environment to ensure it is safe for themselves and others to approach the injured student (e.g., stop ongoing activity, secure equipment).
    - Assess the Injury: Conduct a primary survey to determine the nature and severity of the injury (e.g., check for consciousness, breathing, severe bleeding).
    - Provide Immediate Care: Administer appropriate first aid based on their training (e.g., CPR, use of an AED, controlling bleeding, stabilizing a suspected fracture).
- 2. Emergency Action Plan Activation:
    - Activate EMS: Immediately delegate someone to call 911 or make the call themselves if alone. Provide clear information about the location and nature of the emergency.

- Retrieve Emergency Equipment: Ensure someone retrieves the first aid kit and AED.

- 3. Supervision and Management:
  - Supervise Other Students: Ensure the rest of the class is safe, supervised, and removed from the immediate area to protect the injured student's privacy and allow for care.
  - Direct Assistance: Delegate specific tasks to responsible students or other staff (e.g., "You, go to the office and tell them we need an ambulance at the gym").

- 4. Communication and Documentation:
  - Contact Parents/Guardians: Inform the school administration to contact the student's parents or emergency contacts.
  - Document the Incident: Complete an incident report form as soon as possible after the event, detailing what happened, the care provided, and who was notified.

- 5. Preparation (Pre-event Responsibility):
  - Know the EAP: Be thoroughly familiar with the school's specific Emergency Action Plan, including the location of phones, first aid kits, and AEDs.
  - Maintain Certification: Hold current certifications in First Aid, CPR, and AED use.

21. **Risk Reduction:**

- **Ceiling Height:** A minimum height of 12-14 feet prevents athletes from hitting overhead beams or lights during jumps or Olympic lifts.
- **Flooring:** Shock-absorbent rubber flooring protects against injuries from dropped weights and reduces noise. Non-slip surfaces prevent slips and falls. Specific surfaces like turf are safe for agility work.
- **Supervision Location:** An elevated or centrally located office with a clear line of sight allows coaches to monitor the entire facility for unsafe behavior or accidents.
- **Mirror Placement:** Placed at least 20 inches from the floor to prevent breakage from rolling weights. They also allow for self-monitoring of technique, reducing injury risk from poor form.

- **Electrical Service:** Properly grounded outlets and circuits prevent electrical shocks. Placing outlets strategically avoids trip hazards from cords.
- **Access & Traffic Flow:** Wide, clear walkways (min. 36 inches) and double doors prevent congestion, allow for safe movement of people and equipment, and provide clear paths for emergency egress.
- **Lighting:** Adequate lighting (50-100 lumens) ensures all areas are well-lit, helping athletes see equipment and potential hazards clearly.

B. Training Efficiency:

- **Equipment Zoning:** Grouping equipment by type (free weights, machines, cardio) creates efficient workout zones, minimizes transition time between exercises, and facilitates circuit training.
- **Traffic Flow:** A logical layout prevents congestion and allows athletes to move smoothly between stations, maximizing the use of training time.
- **Environmental Control:** Maintaining a comfortable temperature (68-78°F) and low humidity (<60%) with good air circulation prevents heat-related illnesses and keeps athletes comfortable, enabling them to train harder and longer.
- **Adequate Space per Athlete:** Providing a minimum of 100 sq ft per athlete ensures there is enough room to perform exercises safely and effectively without interference from others.
- **Storage:** A dedicated storage room for extra equipment and cleaning supplies keeps the main training floor organized and uncluttered, creating a more efficient and professional environment.